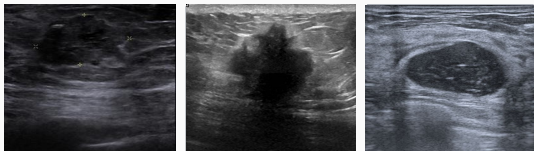


Stage 1: Problem analysis

Where does population bias live in BUS images?



Indian (AIIMS)

Egyptian (BUSI)

Polish (BrEaST)

n = 98 malignant cases per cohort

Radiomic feature analysis

650 features across 7 transform domains

ComBat

scanner
harmonization

**Kruskal–
Wallis**

non-parametric
test

$\epsilon^2 > 0.06$

effect-size
filter

Key finding
60%

of divergent features reside in the wavelet domain

Stage 2: Proposed framework

Selectively harmonize population-specific signal

Input BUS image

Wavelet decomposition

LL, LH, HL, HH subbands

Adaptive Wavelet Normalization

LL

LH

HL

HH

high attention

suppressed

Attention U-Net

backbone

Discriminator

gradient-reversed

**Population-invariant
tumor representation**

Stage 3: Clinical outcomes

Fair performance + clinical transparency

**Consistent segmentation performance
across populations**

Dice > 0.64 across all
leave-one-out configurations

**Interpretable attention coefficients
grounded in ultrasound physics**

Subband weights aligned with
ultrasound physics

GradCAM-based tumor localization

GradCAM reveals anatomically consistent and
uniformly spread activation maps

Equitable, trustworthy deployment